

Belief Coevolution in a Social Network of Generalist and Specialist Large Language Models

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Abstract

CoevolveSim is a network-mediated belief diffusion framework for studying how generalist and specialist LLM-agents update beliefs across social networks. Unlike previous work that relies on behavioral proxies, such as debate summaries or task completion rates, CoevolveSim tracks beliefs through controlled multi-round exchanges. In each round, agents observe their neighbors’ current beliefs before updating their own. Our framework separates the effects of fixed *domain expertise* (the domain-specific LLM behind each agent) from the portrayed *social role* (a label communicated to both the agent and its neighbors). We run 1 280 simulations varying role assignment and LLM heterogeneity across two network structures and 20 medical-indication statements. We find that persona-style prompting yields only marginal changes in individual belief updating and a near-zero shift in final agreement levels. In contrast, using domain-finetuned specialist LLMs alongside generalists yields agreement more than twice that of role assignment alone. It also gives rise to persistent opinion leaders and followers who would otherwise be absent. Our results show that realistic simulation of belief diffusion in multi-agent LLM systems requires a diverse set of LLMs. ¹

1 Introduction

Large Language Models (LLMs) are becoming integral to the global information ecosystem. Nearly two-thirds of companies now use generative artificial intelligence (AI) to produce written content (McKinsey & Company, 2025), with AI-generated text appearing in contexts ranging from UN press releases to scientific articles (Liang et al., 2025a,b). On social knowledge platforms such as Quora, the proportion of AI-generated posts increased from 2% in 2022 to 39% by late 2024 (Sun et al., 2025). Thus, LLMs are no longer merely

external tools applied to information but are participants in how information is produced and circulated (Stanford HAI, 2025).

At the same time, LLMs are increasingly deployed in multi-agent scenarios (Tran et al., 2025), including debates, collaborative reasoning, and assistant teams (Guo et al., 2024; Wu et al., 2024). LLMs are also used to simulate diverse human populations, including persona-prompted voters, patients, and domain experts (Chuang et al., 2024; Taubenfeld et al., 2024). As such, LLMs play two main roles in the information ecosystem: generators of public-facing content and stand-ins for agents with beliefs.

However, we still lack a systematic understanding of *how* these beliefs form and propagate in LLMs. Existing research evaluates multi-agent LLM systems primarily through behavioral proxies,² including task completion, votes, and debate transcripts, rather than examining the underlying belief states that drive these behaviors (Guo et al., 2024). Ye et al. (2026) caution that LLM social simulations have shifted from exploring *what might happen* to making policy-level claims about *why* and *what should happen next*. Ng et al. (2026) further argue that LLM agent states, local interactions, and population-level dynamics remain *jointly* understudied in multi-agent LLM systems. These dimensions correspond to three factors that could shape beliefs in LLM populations: (1) What does an agent “know” from its finetuned training (*domain expertise*)? (2) How does an agent present itself to its peers (*social roles*)? (3) With whom does the agent interact (*network structure*)? We address these with CoevolveSim, an agent-based simulation framework in which generalist and specialist LLMs, embedded in social networks, exchange and update beliefs about factual statements

¹Code and simulation data: carlomarxdk/coevolve-sim.

²By “behavioral proxies,” we refer to observable actions such as the messages LLMs generate and the decisions they make in collaborative tasks.

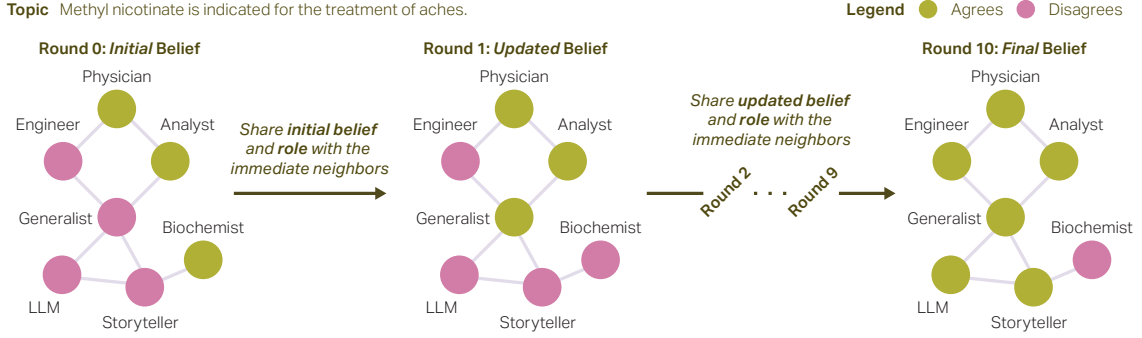


Figure 1: **Schematic overview of CoevolveSim belief diffusion.** Given a discussion topic, CoevolveSim first asks each agent for its initial belief without providing information about the neighbors (Round 0). Then, CoevolveSim facilitates the exchange of beliefs and social roles between agents and their immediate neighbors. In each subsequent round, CoevolveSim evaluates the belief of each agent after providing a summary of the beliefs of its neighbors from the previous round. After T rounds of interaction, agents reach their final belief states. In this illustration and in our experiments, we set $T = 10$. (Note: The statement here is a factually correct statement.)

over synchronous rounds (Figure 1). Our **key contributions** are:

- We introduce CoevolveSim, a multi-agent framework that tracks belief dynamics of LLM agents in social networks.
- We separate the effects of domain expertise and social role on belief dynamics.
- We run 1 280 simulations across four controlled scenarios, two network structures and 20 medical statements.
- We show that model heterogeneity, the use of specialist and generalist LLMs, is the dominant driver of population-level belief change: it gives rise to opinion leaders and followers, and yields an increase in consensus twice that of the persona-style role assignment.
- We find a dissociation between agent- and population-level predictability: in specialist scenarios, surrogate models that account for the beliefs of neighbors and the agent’s identity features improve per-agent transition accuracy, yet only global population-level belief is needed to reproduce collective consensus dynamics.

2 Related Work

Multi-agent LLM populations are often evaluated using behavioral proxies, such as task completion, voting, or debate transcripts (Guo et al., 2024; Wu et al., 2024; Chen et al., 2024), rather than by examining the belief states that drive those behaviors.³ Debate frameworks assess whether a group

of LLM agents converges on factually correct answers (Wang et al., 2023; Khan et al., 2024; Prasad and Nguyen, 2025). Similarly, recent works measure beliefs from sampled texts (Breum et al., 2024; de Wynter and Yuan, 2025; Mannekote et al., 2025) instead of evaluating the underlying belief distributions. Pal et al. (2025) show that when challenged, LLM belief consistency varies across tasks and LLMs, and Bilgin et al. (2025) demonstrate that consensus among LLM agents amplifies persuasion. However, it remains unclear whether these findings generalize to cases when LLM agents are based on different LLMs. Brockers et al. (2025) further highlights that bias decomposition, such as disentangling interaction effects from topics, is *essential* for interpreting LLM opinion dynamics.

LLM Personas and Expertise. Diversity in multi-agent simulations is introduced through prompt-based personas (assigning different social roles to the same underlying LLM) or finetuned LLMs (e.g., with specialization in law). Persona-prompted agents often revert to default model behavior over extended interactions, without the behavioral consistency of finetuned LLMs (Bhandari et al., 2025; Taubenfeld et al., 2024; Villa et al., 2024). Chuang et al. (2024) simulate opinion dynamics with homogeneous persona-prompted LLM agents and find that multi-turn interactions drive consensus, without testing the effect of varying the LLMs that power agents.

Finetuned specialist LLMs, in contrast, show greater robustness on knowledge-intensive tasks (Tanwar et al., 2025; Chen et al., 2023; knowledge.

³We define the LLM-agent’s **belief state** as the probability it assigns to a statement being true, false, or outside of its

Mullick et al., 2024), while populations of identical models are vulnerable to consensus collapse and bias propagation (Breum et al., 2024; Estornell and Liu, 2024; Xiong et al., 2023).

Network Structure and Dynamics. Network structure shapes collective opinion dynamics in humans (Bakshy et al., 2011): local clustering, average path length, and degree heterogeneity influence consensus (DeGroot, 1974; Friedkin and Johnsen, 1990). However, multi-agent LLM simulations rarely consider network structure. Chuang et al. (2024) fix a single network family, and (Gao et al., 2023) treat the underlying network structure as a backdrop rather than a manipulated factor. In contrast, Ng et al. (2026) argue that network structure should serve as a prior in multi-agent AI, since population-level outcomes emerge jointly from LLM agent states and interaction networks.

CoevolveSim addresses these gaps by distinguishing (1) fixed domain expertise in specialized LLMs, (2) portrayed social roles of LLMs, and (3) network structure across four scenarios in networked belief exchange.

3 Belief Diffusion Framework

We propose CoevolveSim, a network-mediated belief diffusion framework (Figure 1). Here, generalist LLMs (without finetuning) and specialist LLMs (finetuned in various fields such as clinical medicine, chemistry, cybersecurity, mathematics, *etc*; see Table A2) act as LLM agents embedded in an undirected social network G .

In each round t , agent a_i receives a template message containing its assigned social role, a discussion statement, and a summary of the beliefs of its neighbors from round $t-1$. Belief updates are synchronous and Markovian: all agents condition on the same previous-round information and use only the latest neighborhood summary. This design allows us to examine how social role, domain expertise, and neighborhood exposure (social network structure) shape beliefs over $T = 10$ rounds.

Next, we define domain expertise and social roles (§3.1); specify network structures (§3.2); formalize belief extraction (§3.3); and define the neighborhood summary (§3.4). Table A1 lists the notation used in this section.

3.1 Domain Expertise and Social Roles

Each LLM agent is characterized by a *domain expertise* and a *social role*. **Domain expertise** re-

flects an LLM’s finetuning domain. For example, Llama3-Med42 (Christophe et al., 2024) is finetuned on medical/clinical text, so we assume it has expertise in medicine. **Social role** is a *social label* assigned to an agent and communicated to both the agent and its neighbors. Social roles function as social identities, such as “clinical physician,” “mathematician,” or “chemist”. In particular, social roles do not necessarily correspond to expertise. An LLM agent could be an expert in medicine but could portray itself as a mathematician. This separation allows us to test whether belief dynamics is driven by what an LLM *knows* (domain expertise) or by what it is *labeled as* (social role).

3.2 Social Networks

We embed n LLM agents in an undirected graph G , where each agent a_i is a node and edges represent social connections. Every agent a_j with an edge to agent a_i is part of a_i ’s neighborhood $\mathcal{N}(i)$. To study how the structure of a network shapes the dynamics of belief, we use two families of random graphs for G : (1) Erdős and Rényi (1959) (ER) and (2) Watts and Strogatz (1998) (WS). These families are described in §4.3.

3.3 Belief Extraction

An LLM agent $\mathcal{M}$ maps an input sequence $\mathbf{x}$ to a distribution $\mathcal{P}_{\mathcal{M}}$. For any token $\tau \in \mathcal{V}_{\mathcal{M}}$, $\mathcal{P}_{\mathcal{M}}(\tau \mid \mathbf{x})$ gives the (uncalibrated) probability that τ continues $\mathbf{x}$. We extract beliefs using template messages $\mathbf{x}$ and record the probabilities of specific tokens (Savcisen and Eliassi-Rad, 2025).

Message. At a round t , LLM agent a_i receives a message $\mathbf{m}_i^{(t)} = (\mathbf{x}_r, \mathbf{x}_s, \mathbf{x}_n^{(t)})$ containing the agent’s role $\mathbf{x}_r$, the discussion statement $\mathbf{x}_s$ and a summary $\mathbf{x}_n^{(t)}$ of neighbors’ beliefs from round $t-1$, with $\mathbf{x}_n^{(0)} = \emptyset$. The message ends with a multiple-choice question asking the agent to classify the statement $\mathbf{x}_s$ as (1) *correct*, (2) *incorrect*, or (3) *neither*. See Figure 2.

Belief. Given message $\mathbf{m}_i^{(t)}$, we define LLM agent i ’s belief at round t as

$$\mathbf{b}_i^{(t)}(y \mid \mathbf{m}_i^{(t)}) = \mathcal{P}_{\mathcal{M}}([k_y] \mid \mathbf{m}_i^{(t)}). \quad (1)$$

where $k_y \in \{1, 2, 3\}$ indexes $y \in \{\text{correct, incorrect, neither}\}$. The next-token probability assigned to the corresponding option is the *belief score*. Additionally, any residual probability mass not captured by these tokens is added to

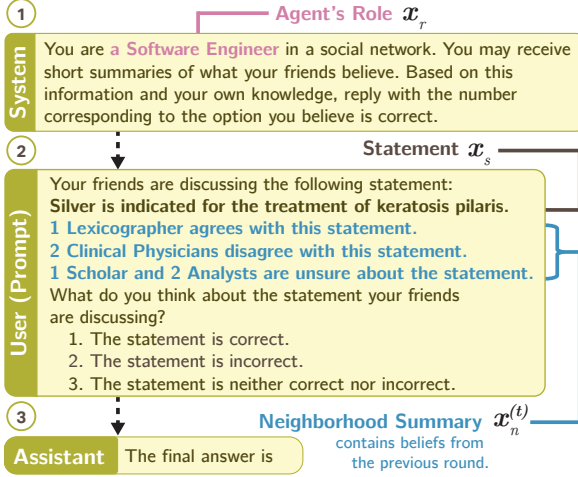


Figure 2: **Message structure for belief probing.** Each LLM agent receives a message that follows the Hugging Face chat template: (1) a *system* prompt assigning the agent’s role x_r and task instructions; (2) a *user* prompt containing the statement x_s , the neighborhood summary $x_n^{(t)}$ ($x_n^0 = \emptyset$), and the multiple-choice question; and (3) the *assistant* prompt from which we extract the next-token probabilities to obtain belief scores.

$b_i^{(t)}$ (*neither*) so that the three scores sum to 1.⁴ The LLM agent’s *discretized belief* is

$$B_i^{(t)} = \arg \max_y b_i^{(t)}(y). \quad (2)$$

3.4 Neighborhood Summary

At round t , we collect the previous round’s discretized beliefs $B_j^{(t-1)}$ from a_i ’s neighbors and group the neighbors into three categories: *agree*, *disagree*, and *unsure*. For each group, we generate a sentence listing the corresponding neighbor roles and counts (see Figure 2), and concatenate the three sentences to form a summary of the neighbors’ beliefs $x_n^{(t)}$. No neighborhood summary is provided at $t = 0$.

To reduce ordering bias, we randomly shuffle social roles within each sentence and randomize sentence ordering before inserting them into the template. Thus, the summary includes only the beliefs of the previous round, and agents have no access to previous rounds.

4 Experimental Setup

In this section, we describe the large language model used in our study (§4.1), discussion statements (§4.2), network generation (§4.3), simu-

lation scenarios (§4.4), and evaluation strategies (§4.5).

4.1 Large Language Models

To study how domain expertise is associated with belief formation, we use 15 LLMs from the Llama-3 (Grattafiori et al., 2024) family. Llama-3.1-8B-Instruct serves as the *generalist*. The other 14 LLMs serve as *specialists* and are finetuned on specific domain corpora, such as biomedical texts, cybersecurity reports, programming tasks, etc. Each LLM has approximately 8 billion parameters, so behavioral differences result from finetuning rather than model size. Table A2 provides more information about the LLMs used in our experiments.

4.2 Discussion Statements

We sample affirmative medical-indication statements referencing real-world entities from the corpus introduced by Savcisen and Eliassi-Rad (2025). An example statement is “Terbutaline is indicated for the treatment of asthma.”

We use zero-shot predictions from a medical LLM (namely, Llama3-Med42 (Christophe et al., 2024)) and 14 additional LLMs to construct features that capture the (1) binary ground-truth label of a statement (*true*, *false*), (2) label predicted by the medical LLM, (3) accuracy of the medical LLM, (4) average accuracy of other LLMs, (5) agreement between the medical LLM and other LLMs. We apply the maximin criterion (Johnson et al., 1990) to select 20 diverse statements. In this way, we maximize the minimum pairwise distance in this feature space. We provide additional details and the full statement list in Appendix C and Table A4.

4.3 Social Networks

We represent social connections between agents via two network models: (1) Erdős and Rényi (1959) (ER), and (2) Watts and Strogatz (1998) (WS).

In ER, $G(n, p)$, each pair of agents is connected with probability p . It produces locally tree-like networks without tight clusters or heavy-tailed distributions (Barabási and Pósfai, 2016). In WS, $G(n, \beta, k)$, the network is constructed by rewiring edges of a k -regular ring lattice with probability β . It produces high local clustering with short average path lengths.

We fix $n = 48$ for both networks structures, with $p = 0.3$ for ER (Leskovec, 2017) and $k = 8$, $\beta =$

⁴The $b_i^{(t)}$ values are uncalibrated probabilities.

0.1 for WS. We select 8 network instances using the maximin criterion described in Appendix C.2 for each network structure and refer to these as network realizations.

4.4 Scenarios and Simulations

We examine belief diffusion across *four* scenarios. These scenarios isolate the effects of expertise, role assignment, and role-expertise alignment on belief diffusion.

I. Baseline Generalists. All agents use the same generalist LLM and receive the same generic LLM role, serving as a no-expertise, no-role baseline.

II. Generalists with Random Roles. All agents use the same generalist LLM, but each is assigned a random role drawn from Table A2. This scenario isolates the effect of role information alone.

III. Specialists with Random Roles. Agents use finetuned specialist LLMs with randomly assigned roles. The list of LLMs and roles is provided in Table A2. This scenario separates model expertise from role assignment.

IV. Specialists with Matching Roles. Each agent is a finetuned specialist LLM assigned a role that matches its expertise. Compared with Scenario III, this scenario isolates the marginal effect of role-expertise alignment.

Runs. Each scenario is simulated multiple times; we refer to each individual simulation as a *run*. Each run consists of $T = 10$ rounds, where round 0 collects agents' initial beliefs before introducing neighbor information. In total, we perform 1280 runs: 4 scenarios, 2 network structure with 8 realizations each, and 20 discussion statements.

4.5 Evaluation

Our analysis proceeds in two stages. We first characterize belief trajectories and compare dynamics across scenarios and network structures (§4.5.1). We then evaluate the extent to which classical opinion-dynamics models explain the observed LLM belief dynamics using a hierarchy of increasingly expressive surrogate models (§4.5.2).

4.5.1 Dynamics Analysis

We characterize belief trajectories along three agent-level aspects: magnitude of change (*Plasticity*), directional consistency (*Directedness*), and Outgoing *Influence*. Additionally, we consider a population-level outcome – *Consensus*. First, we define agent a_i 's per-round belief change as: $\Delta \mathbf{b}_i^{(t+1)} = \mathbf{b}_i^{(t+1)} - \mathbf{b}_i^{(t)}$.

Plasticity is the average magnitude of an agent's belief change per round (across a run):

$$\text{Plasticity}(i) = \frac{1}{T-1} \sum_{t=0}^{T-1} \frac{1}{2} \left\| \Delta \mathbf{b}_i^{(t+1)} \right\|_1 \quad (3)$$

Agents with high plasticity update frequently, while agents with low plasticity remain stable.

Directedness captures whether an agent's belief moves in a consistent direction:

$$\text{Directedness}(i) = \frac{\left\| \mathbf{b}_i^{(T)} - \mathbf{b}_i^{(0)} \right\|_1}{\sum_{t=0}^{T-1} \left\| \mathbf{b}_i^{(t+1)} - \mathbf{b}_i^{(t)} \right\|_1} \quad (4)$$

Directedness closer to 1 indicates that an agent's beliefs change in a single direction across rounds, while 0 indicates belief oscillation.

Influence measures the average magnitude of belief changes in an agent's neighbors following the agent's own update.

$$\text{Influence}(i) = \text{avg}_{t, j \in \mathcal{N}(i)} \left\| \Delta \mathbf{b}_j^{(t+1)} \right\|_1 \quad (5)$$

A higher influence indicates that belief updates in a_i are associated with greater subsequent updates among its neighbors.

Finally, **consensus** measures the agreement between the discretized beliefs of all agents:

$$\text{Cons}(t) = \frac{1}{n(n-1)} \sum_{i \neq j} \mathbb{1} \left[B_i^{(t)} = B_j^{(t)} \right]. \quad (6)$$

In our analysis, we look at the difference in consensus between initial ($t = 0$) and final rounds.

Statistical Inference. We fit linear mixed-effects models with fixed effects for scenarios (I–IV) and the ER and WS networks. We add random intercepts for each statement and network realization. This design is applied to *plasticity*, *directedness*, and *change in consensus*. We report estimated marginal means (EMMs) with 95% confidence intervals in §5.

Variance decomposition. To assess if certain agents exert more influence than others, we fit a variance-components model (Scheipl et al., 2008) to agent-level outgoing $\text{Influence}(i)$. We add random intercepts for agent, network realization, and statement. This lets us quantify how much variance in *Influence* is attributable to agent identity, relative to network realization and statement. For each

random-effects component, we report the intraclass correlation coefficient (ICC), which is the proportion of total variance attributable to that source. A higher agent-level ICC suggests the presence of persistent opinion leaders.

We provide full model specifications and additional contrast analyses in Appendix D.

4.5.2 Analysis with Surrogate Models

To assess whether the observed belief dynamics can be explained by classical opinion-dynamics mechanisms, we fit a hierarchy of multinomial logistic transition models, which we refer to as *surrogate models*. The hierarchy adds information incrementally: (i) individual persistence, (ii) global population belief, (iii) local neighborhood belief, and (iv) agent identity signals, namely the underlying LLM and assigned role. This structure provides a controlled ablation, enabling us to test whether each additional information source improves our ability to capture belief transitions. Formally, each model estimates

$$\Pr(B_i^{(t+1)} = y \mid \phi_i^{(t)}), \quad (7)$$

where $\phi_i^{(t)}$ denotes the transition-model’s feature vector at round t , and $B_i^{(t)}$ is the discretized belief of agent a_i .

M1: Individual Persistence. Conditions only on a_i ’s current belief $B_i^{(t)}$ (one-step self-transition dynamics).

M2: M1 + Global drift. Adds global belief aggregates, inspired by DeGroot (1974).

M3: M1 + Local influence. Uses persistence with the neighborhood belief aggregate, analogous to the local social influence in Friedkin and Johnsen (1990).

M4: Local social influence + Identity. Extends M3 with categorical features for each agent’s underlying expertise and assigned social role.

We evaluate the surrogate hierarchy in two ways. First, we assess model fit via accuracy in capturing held-out transitions. Second, we treat the fitted models as generative surrogates and examine whether simulated trajectories reproduce the empirical final-round consensus and per-agent belief labels; see §5. Full model specifications and rollout evaluations are provided in Appendix E.

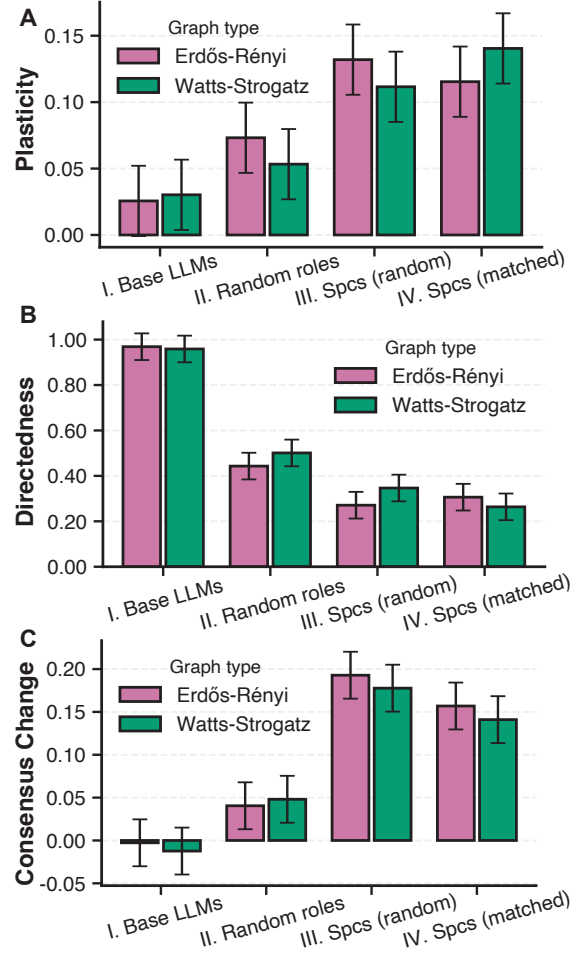


Figure 3: **Belief dynamics across settings and network structures.** Estimated marginal means (95% CI) from mixed-effects models. *Spcs* stands for *specialists*. (A) *Plasticity*: average per-round magnitude of belief change. Increases from baseline (I) through random specialists (III); role-alignment (IV) reduces plasticity in ER but increases it in WS. (B) *Directedness*: consistency of belief updates across rounds. Drops sharply once roles are introduced (I→II) and reaches its minimum in WS with matched specialists (IV). (C) *Consensus change*: shift in overall belief agreement between the initial and last rounds. Model heterogeneity drives the largest shifts in consensus, while role assignment and heterogeneity shape agent-level dynamics.

5 Results

We first examine how belief dynamics differ across scenarios and network structures. We study plasticity, directedness, and consensus change (§5.1). Next, we assess whether surrogate opinion-dynamics models can reproduce the observed trajectories (§5.2).

5.1 Dynamics

Plasticity is lowest in the baseline scenario (I) under both structures, with estimated means of 0.02 in ER and 0.03 in WS (Figure 3.A). Assigning random roles to generalist agents (II) increases plasticity to 0.07 in ER and 0.05 in WS. Plasticity increases further in the random-specialists setting (III), while changes in role-specialization matching (IV) are network-dependent: in ER, plasticity decreases relative to random specialists (0.13 to 0.12), whereas it increases in WS.

Directedness follows the opposite pattern (Figure 3.B). It is highest in the baseline setting (I): few updates occur, but those that do are monotone. Introducing random roles (II) produces the largest drop in directedness, to ≈ 0.44 in ER and 0.50 in WS. Directedness decreases further in the specialist settings (III–IV), with the lowest values observed in WS under matched specialists (IV).

Consensus change (Figure 3.C) is minimal in the baseline scenario (I) and small under random roles (II). The largest shifts occur in scenario III, reaching 0.19 in ER and 0.18 in WS. Matching roles to specialization (IV) attenuates these shifts to 0.16 in ER and 0.14 in WS.

Outgoing influence differs mainly in the specialist settings (III–IV) compared to the baseline scenarios (I–II), as shown in Figure 4. That is, the agent-level ICC is near zero in I–II, indicating no differences in influence across agents. In specialist scenarios, both agent- and network-level ICCs increase, underscoring that network structure matters more when agents are heterogeneous. Random specialists (III) show about twice the agent-level variance of matched specialists (IV), highlighting an additional contrast within the specialist cases.

Additional contrast analyses in Appendix D confirm the following key findings: model heterogeneity (III–IV) is the strongest driver of consensus change, role and expertise reduce directedness, and role-alignment effects are smaller and depend on the network structure.

5.2 Surrogate Modeling

We evaluate the surrogate models across two criteria. First, we consider one-step predictive accuracy on held-out transitions, evaluated using the final-state Matthews correlation coefficient (MCC). Second, we measure consensus agreement between simulated and empirical runs, which we refer to as *consensus fidelity* (see Appendix E for details).

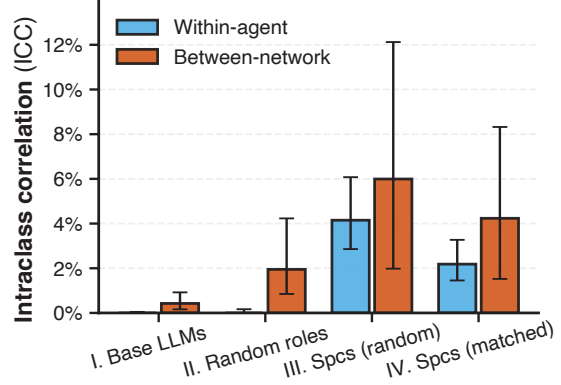


Figure 4: **Variance decomposition of outgoing influence.** Intraclass correlations ($\pm 95\%$ CI) from mixed-effects models with random intercepts for agent, graph, and statement. *Spcs* stands for *specialists*. **Within-agent** measures the proportion of total influence variance attributable to agent-level differences; **between-network** measures variance attributable to graph realizations. Statement and residual components are not shown. Within-agent ICC is statistically zero in homogeneous-model scenarios (I, II) and rises to 2–4% in scenarios with specialist models (III, IV). Agent-level differences in outgoing influence emerge only when finetuned LLMs are present.

Figure 5 shows that final-state MCC depends both on setting and networks structure. In the homogeneous settings (I–II), MCC is stable across surrogate models, with ER generally outperforming WS. In the specialist settings (III–IV), additional model complexity yields modest gains in MCC, especially for WS.

Figure 6 shows that consensus fidelity exhibits a different pattern. In the homogeneous settings (I–II), the simplest models, M1 and M2, achieve the best fidelity, while M3 and M4 underperform despite their higher final-state MCC. In the specialist settings (III–IV), adding global information yields a substantial gain in fidelity. However, if we account only for local neighborhood composition (M3) and agent identity (M4), we do not get any significant improvements. Thus, strong one-step predictive performance does not necessarily translate into accurate reproduction of population-level consensus dynamics.

6 Discussion

Our results identify three mechanisms that shape belief diffusion in networks of LLM agents: role assignment, model expertise, and their interaction with network structure.

Role labels reshape how agents update, but

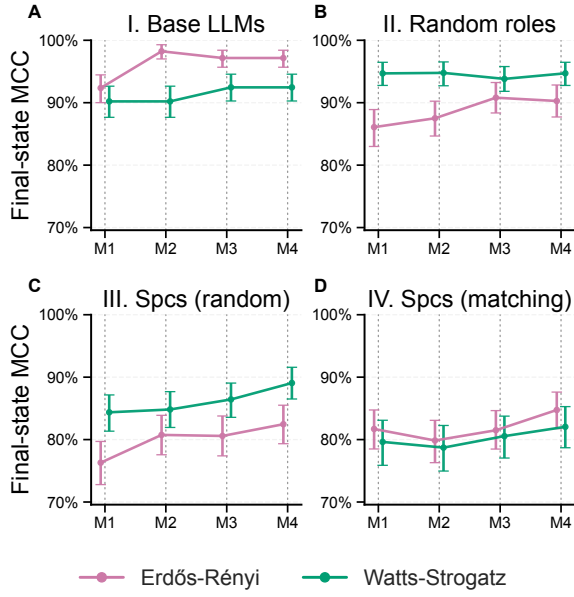


Figure 5: **Final-state MCC** measures the accuracy of the surrogate’s predicted final belief against ground truth. *Spcs* stands for *specialists*. Each panel corresponds to one of four experimental settings: (A) Base LLMs, (B) Random roles, (C) Specialists with random roles, and (D) Specialists with matched roles. Each line connects point estimates (95% bootstrap CIs) across surrogate models of increasing complexity for ER and WS networks. Surrogate models with additional social information provide limited benefit in homogeneous settings, but modestly improve predictive accuracy in specialist-driven settings.

not where they converge. Assigning roles produces a sharp reduction in directedness ($d = -0.49$) with a small increase in plasticity ($d = 0.04$) and yields only marginal consensus shifts ($d = 0.05$). Role labels primarily alter the direction of belief updating rather than its frequency: updates become less monotone and more oscillatory. This is consistent with Li et al. (2025), who find that persona-prompted agents produce surface-level expert language without behavioral consistency. In persona-based multi-agent systems, role diversity alone does not produce meaningful divergence in collective outcomes.

Model heterogeneity is the dominant driver of population-level change. Consensus shifts emerge only when specialist models are present (III, IV), with the model effect on consensus change more than twice that of scenario II ($d = 0.14$ vs. $d = 0.05$). The variance decomposition reinforces this: agent-level ICC for outgoing influence is statistically zero in homogeneous scenarios and rises only with specialist models.

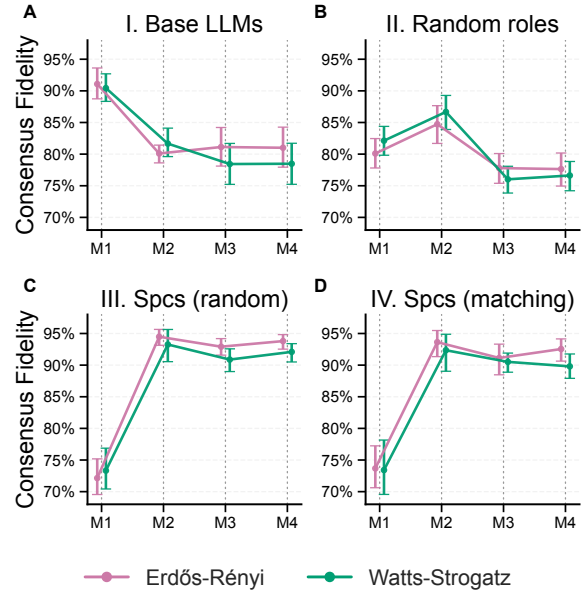


Figure 6: **Consensus Fidelity for surrogate model evaluations.** *Spcs* stands for *specialists*. Each panel corresponds to one of four experimental settings (A) Base LLMs, (B) Random roles, (C) Specialists with random roles, (D) Specialists with matched roles, and each line connects point estimates (95% bootstrap CIs) across surrogate models of increasing complexity for Erdős-Rényi and Watts-Strogatz networks. In homogeneous settings, simpler surrogates achieve the highest consensus fidelity, while specialist settings require at least global social information to reproduce empirical consensus dynamics.

Populations of homogeneous LLMs exhibit low plasticity and limited capacity for collective belief revision; diversity of the underlying model, not merely persona, appears necessary for LLM-agents to propagate information differentially.

Role-expertise alignment interacts with network structure. Aligning roles with expertise (IV) has opposite effects across network structures: in ER, it reduces plasticity and slightly increases directedness, while in WS, it increases plasticity and lowers directedness. The mechanism behind this reversal is not tested in the current study. One possible explanation is that WS’s high local clustering amplifies alignment signals within tightly connected neighborhoods, inducing more oscillatory updating.

One-step predictive accuracy does not guarantee realistic collective dynamics. In homogeneous settings (I, II), belief transitions are well captured by persistence and global information alone; adding local information (M3) and identity (M4) leaves final-state MCC almost unchanged but de-

grades consensus fidelity. This suggests that additional local complexity may overfit short-term transitions, with small errors compounding over simulated trajectories. In specialist settings (III, IV), incorporating social information is necessary to recover consensus dynamics, yet detailed local and identity features continue to improve per-agent fit without improving consensus fidelity.

7 Conclusion

Current multi-agent LLM studies often track observable behavior, but not the belief dynamics that underlie it. This work introduces CoevolveSim and identifies three distinct mechanisms that shape belief diffusion within networks of large language model (LLM) agents. The following findings are grounded in a controlled setting using Llama-based models across 20 medical-indication statements

Role assignment slightly increases plasticity and substantially reduces directedness, yet leads to only minimal changes in consensus. Finetuned specialist models drive population-level dynamics and produce detectable asymmetries in influence: the emergence of leaders and followers. Belief change is primarily mediated by the underlying finetuned model rather than by assigned roles. Network topology appears to matter more in specialist settings: the dynamics under the two topologies start to diverge.

Finally, we demonstrate a dissociation between agent- and population-predictability: baseline settings are sufficiently captured by persistence and global drift. In specialist settings, global information is necessary to recover collective dynamics, whereas local neighborhood and identity improve per-agent fit without translating into better population-level predictions.

These results show that a faithful simulation of diverse belief dynamics in multi-agent LLM systems requires model-level heterogeneity; persona prompting alone is insufficient.

Limitations

Scope of the experiments Our study deliberately trades *breadth* for *control*, and several design choices limit the generalizability of our conclusions. Statements are chosen from a single domain (affirmative medical indications), which keeps the ground truth verifiable but leaves open how dynamics would change for contested, vague, or compositional claims.

The LLM agent pool is narrow by design to better control for confounding factors such as family or scale. Similarly, our study does not separate the effects of finetuning data from those of optimization choices. We examine small populations ($n = 48$) over a short timeline ($T = 10$ rounds), which is sufficient to identify phase-transition-like cascades but insufficient to capture slow drifts or behavior at realistic community scales.

Interaction protocol. We deliberately use minimal update rules. Our protocol is synchronous, Markovian, and broadcast-style. Each LLM agent conditions only on the neighborhood summary from the previous round, not the entire interaction history. This design isolates the effects of social influence from memory and dialogue. However, it excludes practical mechanisms such as asynchronous turn-taking, persistent context, free-form messaging, and persuasion effects.

Belief inference and causality scope. We read beliefs from next-token probability over a small discrete label set. This produces a tractable, comparable signal across heterogeneous models, but it cannot characterize what happens inside an LLM (its inner representations) or address its sensitivity to prompt structure.

Representation-based probes of latent truth or epistemic stability would complement this output view. These could help disentangle genuine belief revision from response conformity. Our causal claims are similarly bounded: the dynamics analysis detects role, model, alignment, and heterogeneity effects across our four scenarios. However, we do not use formal mediation analysis or targeted interventions to localize the mechanism. No human participants are involved. Embedding humans into the same framework and comparing their update behavior with that of LLM agents under identical conditions is a natural next step.

Ethical Considerations

CoevolveSim is a simulation of LLM-agent belief dynamics, not a study of human subjects. Therefore, the results should not be interpreted as evidence about human behavior or used as a basis for human-facing policy claims. Our findings are limited to a single Llama-3 family and may not generalize to other model families, sizes, languages, or cultural settings.

Use of AI. We used (1) Claude Code and GitHub Copilot for code annotations and formatting assistance and (2) Grammarly for grammar and style support. After generating the manuscript, we asked Claude Code to proofread and review it and made updates as we considered appropriate.

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A Notations and Abbreviations

Symbol	Description	Notes
$\mathcal{M}$	Large language model	
$\mathcal{V}$	Vocabulary of the model $\mathcal{M}$, consists of tokens	$[\tau_1, \dots, \tau_{ \mathcal{V} }] \in \mathcal{V}$
τ	Token from the vocabulary $\mathcal{V}_{\mathcal{M}}$	e.g. tokens [1], [2]
$\mathcal{P}_{\mathcal{M}}(\tau \mid \mathbf{x})$	Output of $\mathcal{M}$: a conditional probability distribution on tokens	
Agents and Networks		
$G(n, p)$	Erdős–Rényi (ER) random graph model	
$G(n, \beta, k)$	Watts–Strogatz (WS) random graph model	
n	Number of nodes in graph G	
p	Probability of an edge between two nodes in ER graphs	
β	Edge rewiring probability in WS graph	
k	Size of the regular ring lattice in WS graph	
a_i	i -th agent/node in a graph G	
$\mathcal{N}(i)$	Neighbors of a node i	
Belief		
$\mathbf{b}_i^{(t)}$	(Uncalibrated) belief scores of a model $\mathcal{M}$ at a round t	$\sum_y \mathbf{b}_{\mathcal{M}}^{(t)}(y \text{input}) = 1$
$B_i^{(t)}$	Discretized belief label of the i th agent	$B_i^{(t)} = \arg \max_y \mathbf{b}_i(y \mid \mathbf{m}_i^{(t)})$
$b_i^{(r,t)}$	Belief (label) of agent a_i at round t in run r	
$\hat{b}_i^{(t+1)}$	Predicted next-round belief for agent a_i	
T	Total number of rounds	$T = 10$
r	Simulation run index	
Pr	Probability given by surrogate models	
$\mathbf{g}^{(r,t)}$	Global belief-state composition at round t in run r	$(f_{-1}^{(r,t)}, f_0^{(r,t)}, f_1^{(r,t)})$
$\mathbf{n}_i^{(r,t)}$	Neighborhood belief-state composition for agent a_i	$(n_{i,-1}^{(r,t)}, n_{i,0}^{(r,t)}, n_{i,1}^{(r,t)})$
$\mathcal{M}_i$	LLM assigned to agent a_i	
ρ_i	Role assigned to agent a_i	
Inputs and Text		
$\mathbf{m}_i^{(t)}$	Message to i -th agent at a round t	$\mathbf{m}_i^{(t)} = (\mathbf{x}_s, \mathbf{x}_r, \mathbf{x}_n^{(t)})$
$\mathbf{x}_r$	Role specification	Table A2
$\mathbf{x}_s$	Statement (topic of discussion)	Table A4
$\mathbf{x}_n^{(t)}$	Summary of neighbors' beliefs at round t	$\mathbf{x}_n^0 = \emptyset$
$\phi_i^{(t)}$	Feature vector used to predict the next-round belief of agent a_i	For surrogate models
y	Ground-truth label for the statement $\mathbf{x}_s$	

Table A1: Notations used throughout the paper. Symbols are grouped by category.

B Selection of Large Language Models

We select Large Language Models according to the following criteria:

1. The finetuned LLM must be based on Llama-3-Instruct or Llama-3.1-Instruct (Grattafiori et al., 2024).
2. The finetuning must use conversational or instruction-based datasets to maintain the LLM’s chat-like capabilities.
3. The LLM must be publicly available on the [Hugging Face](#) platform.
4. The developers must provide documentation describing the data sources (e.g., URLs and data cards).
5. The model card or associated manuscript must report benchmark results demonstrating that the finetuning yields comparable or improved performance relative to the chat-based Llama-3 model (for a domain-specific task).

Model Name	Specialization (Domain)	Social Role	Developer & Source
Llama-3.1-8B-instruct	Generalist (<i>Base Model</i>)	LLM or Human Participant	Grattafiori et al. (2024)
Bio-Medical-Llama-3-8B	Biomedical Research	Biomedical Researcher	Contact Doctor (2024)
Llama3-Med42-8B	Clinical Medicine	Clinical Physician	Christophe et al. (2024)
Llama-3.1 OpenScholar-8B	Scientific Literature	Academic Scholar	Asai et al. (2026)
Llama3Dictionary-merge	Lexicography / Definitions	Lexicographer	Periti et al. (2024)
Llama-3.1-Hawkish-8B	Financial Markets	Financial Analyst	mukaj (2024)
Llama3-KALE-LM-Chem-1.5-8B	Chemistry	Chemist	Dai et al. (2024)
Foundation-Sec-8B-Instruct	Cybersecurity	Security Analyst	FDTN AI (2024)
UserLM-8b	Social Conversation	Online Friend	Naous et al. (2026)
OpenMath2-Llama3.1-8B	Mathematics	Mathematician	Toshniwal et al. (2024)
Human-Like-LLama3-8B-Instruct	Human-like Conversation	Assistant	Çalk and Akkuş (2025)
Hermes-3-Llama-3.1-8B	Strategic Reasoning	Strategic Planner	Teknium et al. (2024)
suzume-llama-3-8B-multilingual	Multilingual Dialogue	Language Mediator	Devine (2024)
Code-Llama-3-8B	Coding and Maths	Software Engineer	Bawase (2024)
Roleplay-Llama-3-8B	Roleplay / Storytelling	Storyteller	Gallego (2024)

Table A2: Summary of the 15 LLMs used as agents in our study. Each model is characterized by its specialization (i.e., the datasets on which it was finetuned), the social role assigned to it within the experiments, and its developer or source. The selection includes both general-purpose (a.k.a. generalist) and finetuned (a.k.a. specialist) Llama-3 or 3.1. *Note:* Although the generalist LLM can take the social role of a *Human Participant*, this designation refers only to its social identity in the experiment and does not imply human-like qualities. Also note that in the *baseline* experiments, social roles may be assigned at random and thus do not necessarily correspond to the expertise/specialization categories shown here.

C Sampling with Maximin Criterion

The design space of our experiments requires around 10^5 simulations (statements $\times$ network seeds $\times$ settings $\times$ graph types), which is infeasible given the time and resources. To reduce the number of simulations, while ensuring that we select a diverse set of settings (statements and network seeds), we use **maximin criterion** (Johnson et al., 1990). This appendix describes the algorithm, its application in our paper, and the diagnostics we use to verify coverage.

C.1 Maximin Criterion

Given a candidate pool $\mathcal{X} = \{x_1, \dots, x_N\} \subset \mathbb{R}^d$ of size N , we select a subset $S^* \subseteq \mathcal{X}$ of size K whose minimum pairwise Euclidean distance is as large as possible:

$$S^* = \arg \max_{\substack{S \subseteq \mathcal{X} \\ |S|=K}} d_{\min}(S), \quad (\text{A1})$$

$$d_{\min}(S) := \min_{\substack{x, y \in S \\ x \neq y}} \|x - y\|_2. \quad (\text{A2})$$

We use the farthest-point greedy algorithm of Gonzalez (1985) to perform the iterative selection in Algorithm 1.

Procedure. We start by standardizing features of $\mathcal{X}$ to zero mean and unit variance. We then either select the sample with the largest norm, or provide an initial set S_0 . We iteratively select the candidates from $\mathcal{X}$ whose pairwise minimum distance to the already-selected set S^* is largest.

Algorithm 1 Greedy maximin selection

Require: Pool $\mathcal{X} = \{x_1, \dots, x_N\} \subset \mathbb{R}^d$, target size K , optional starting set S_0 .

Ensure: $S \subseteq \mathcal{X}$ approximating S^* in (A1).

```

1:  $\mathcal{X} \leftarrow \text{standardize}(\mathcal{X})$ 
2: if  $S_0$  is non-empty then
3:    $S \leftarrow S_0$ 
4: else
5:    $S \leftarrow \{ \arg \max_{x \in \mathcal{X}} \|x\|_2 \}$ 
6: end if
7: while  $|S| < K$  do
8:   for each  $x \in \mathcal{X} \setminus S$  do
9:      $d(x) \leftarrow \min_{x' \in S} \|x - x'\|_2$ 
10:  end for
11:   $S \leftarrow S \cup \{ \arg \max_{x \in \mathcal{X} \setminus S} d(x) \}$ 
12: end while
13: return  $S$ 
```

C.2 Sampling Network Realizations with Maximin Criterion

Seed	$\bar{\ell}$	C	$\bar{k}$	σ_k	k_0
<i>Erdős–Rényi</i> ($n = 48, p = 0.3$)					
2012155723	1.755	0.233	12.375	2.833	11
2292123236	1.666	0.353	15.750	3.388	20
43019216	1.734	0.322	13.542	3.937	7
1982018573	1.697	0.295	14.458	2.273	11
3781092945	1.707	0.289	13.958	3.041	27
2701326989	1.686	0.347	15.458	3.974	9
3798279163	1.725	0.284	13.625	3.289	15
601535774	1.675	0.307	15.417	2.457	21
<i>Watts–Strogatz</i> ($n = 48, k = 8, \beta = 0.1$)					
1769674741	2.278	0.418	8.000	1.307	10
365860106	2.615	0.573	8.000	0.500	7
395036430	2.302	0.431	8.000	1.041	5
3830640276	2.352	0.468	8.000	0.645	9
1093773836	2.456	0.497	8.000	1.137	8
1356044289	2.447	0.506	8.000	0.791	6
851350617	2.621	0.538	8.000	0.816	9
4262934205	2.277	0.408	8.000	0.979	8

Table A3: Overview of the structural features of the eight graphs selected per network structure via Algorithm 1. These graphs are generated with the use of our graph generators (ER and WS). Features: ($\bar{\ell}$) mean shortest-path length; (C) global clustering coefficient; ($\bar{k}$) mean degree; (σ_k) standard deviation of the degree distribution; (k_0) degree of node 0.

We consider two network structures: Erdős–Rényi (ER) and Watts–Strogatz (WS). For each structure, we run simulations across multiple network realizations to ensure the findings are not driven by a single sample (i.e., we want a representative set of realizations for each network structure).

Specifically, we generate $N_{\text{pool}} = 1000$ candidate realizations by drawing independent seeds for the corresponding generator while holding the following parameters fixed: $n = 48$ for both network structures, $p = 0.3$ for ER, and $k = 8, \beta = 0.1$ for WS.

For each sampled network realization, we compute five structural features:

1. mean shortest-path length,
2. global clustering coefficient,
3. mean degree,
4. standard deviation of the degree distribution,
5. degree of node 0, the node reserved for the Clinical Physician agent (so that the specialist’s local connectivity varies across replicates).

We discard candidate network realizations with more than one connected component. The resulting per-structure feature matrices are passed to Algorithm 1 with $K = 8$. That is, as a result of the maximin sampling, we get eight network realizations per network structure. Table A3 provides an overview of selected network realizations.

C.3 Sampling Discussion Statements with Maximin Criterion

Our statements are drawn from the medical-indication data introduced in Savcicens and Eliassi-Rad (2025). We restrict selection to non-negated statements that reference real-world entities.⁵ An example of such a statement is “*Heparin is indicated for the treatment of pulmonary embolism.*”

Before applying the maximin criterion, we compute zero-shot predictions for each statement s from the medical specialist (Llama3-Med42-8B) and from 15 other LLMs in Table A2. In addition, we define the following features for each statement s :

1. The binary ground-truth label:
 $y \in \{\text{FALSE}, \text{TRUE}\}$ or $\{0, 1\}$, accordingly,
2. Label predicted by a medical specialist.
3. Medical specialist’s accuracy, based on next-token probabilities: $1 - |y - p_{\text{doc}}(\text{TRUE} \mid s)|$.
4. Label-level average accuracy of other LLMs
5. Consensus between medical specialist and other models = $1 - |p_{\text{doc}}(\text{TRUE} \mid s) - \tilde{p}_{\text{other}}(\text{TRUE} \mid s)|$, where $\tilde{p}_{\text{other}}$ denotes the median probability assigned by the 15 other models.

The first two features are binary; the remaining three are continuous in the range $[0, 1]$. Note that feature #4 is computed from the binary label predictions, whereas feature #5 compares probability distributions, so the two are not redundant.

Since we want to have an equal number of true and false statements, we added a constraint to Algorithm 1 that enforces a $K/2$ quota per ground-truth label (we set $K = 20$). That is, we ensure we select 10 true and 10 false statements. Table A4 displays selected statements with the corresponding features.

⁵We exclude neither-valued statements with synthetic entities.

ID	Statement	$\hat{y}_{\text{doc}}$	a_{doc}	a_{other}	c
<i>False statements ($y = 0$)</i>					
false_0	Silver is indicated for the treatment of keratosis pilaris.	0	0.999	0.800	0.948
false_1	Terbutaline is indicated for the treatment of cramps.	1	0.078	0.067	0.966
false_2	Isosorbide is indicated for the treatment of ischemia.	1	0.012	0.867	0.024
false_3	Atorvastatin is indicated for the treatment of aggression.	0	1.000	0.000	0.022
false_4	Manidipine is indicated for the treatment of septic.	0	1.000	0.333	0.554
false_5	VLTS-589 is indicated for the treatment of diabetic macular edema.	1	0.383	0.867	0.386
false_6	Nandrolone decanoate is indicated for the treatment of Duchenne muscular dystrophy.	0	1.000	0.667	0.608
false_7	Vitamin E is indicated for the treatment of pulmonary coccidioidomycosis.	0	1.000	0.267	0.191
false_8	Tetracycline is indicated for the treatment of throat.	0	0.755	0.933	0.763
false_9	Ethinylestradiol is indicated for the treatment of dry cough.	0	1.000	0.067	0.301
<i>True statements ($y = 1$)</i>					
true_0	Methyl nicotinate is indicated for the treatment of aches.	1	0.773	0.933	0.869
true_1	Terbutaline is indicated for the treatment of asthma.	0	0.000	0.133	0.986
true_2	Warfarin is indicated for the treatment of pulmonary embolism.	1	1.000	0.200	0.006
true_3	Tetracycline is indicated for the treatment of Q fever.	0	0.047	0.933	0.188
true_4	Theophylline is indicated for the treatment of asthma.	1	0.531	0.267	0.773
true_5	Sildenafil is indicated for the treatment of pulmonary hypertension.	1	1.000	0.533	0.510
true_6	Polymyxin B is indicated for the treatment of infections of the urinary tract.	0	0.438	0.133	0.571
true_7	SNS-314 is indicated for the treatment of tumors.	0	0.016	0.533	0.589
true_8	Valomaciclovir is indicated for the treatment of viral infection.	0	0.318	0.400	0.920
true_9	Linezolid is indicated for the treatment of nosocomial pneumonia.	1	0.703	0.200	0.307

Table A4: The 20 statements selected via Algorithm 1, with four features. The ground-truth label of the statement (*true*, *false*): we separate false ($y = 0$) and true statements ($y = 1$) into subtables. The features are: ($\hat{y}_{\text{doc}}$) medical specialist’s predicted label; (a_{doc}) medical specialist accuracy; (a_{other}) average accuracy of the 15 other LLMs; (c) consensus score between the medical specialist and the 15 other LLMs, $1 - |p_{\text{doc}}(\text{TRUE} | s) - \tilde{p}_{\text{other}}(\text{TRUE} | s)|$, where $\tilde{p}_{\text{other}}$ denotes the median probability of 15 non-medical LLMs. Within each subtable (*true* or *false*), statements where $\hat{y}_{\text{doc}}$ matches the subtable header indicate the medical specialist agreed with ground truth.

D Additional Details on Descriptive Analysis

This section expands on the methodology described in §3.

D.1 Linear Mixed-Effects Models

Estimated marginal means (EMMs), or least-squares means, are model-based predicted means adjusted for the other terms in the mixed model. To estimate the coefficients, EMMs and confidence intervals, we fit the following linear mixed-effects model for plasticity, directedness, consensus (and consensus change), and outgoing influence (see §4.5.1 for definitions):

`<var-name> ~ scenario * network-type + (1 | statement) + (1 | network-type/seed)`

The random effects structure includes `(1 | statement)` to account for repeated measures across statements and `(1 | network-type/seed)` to capture variation due to different realizations of each network type.

D.2 Variance Decomposition

Scenario	Agent ICC (95% CI)			Network ICC (95% CI)			RLRT <i>p</i>
	ICC	Lower	Upper	ICC	Lower	Upper	
I. Baseline	0.000	0.000	0.001	0.004	0.002	0.009	1.00
II. Random Roles	0.000	0.000	0.002	0.020	0.009	0.042	1.00
III. Random Specialists	0.042	0.029	0.061	0.050	0.020	0.121	0.00
IV. Matched Specialists	0.022	0.015	0.033	0.042	0.015	0.083	0.00

Table A5: Variance decomposition of outgoing influence by scenario, showing agent-level and network-level intraclass correlations (ICC) with likelihood-ratio test results (RLRT). The variation in agents’ influence (leaders and followers) appears only when heterogeneous LLMs are introduced.

To assess whether some agents exert more influence than others, we fit a variance-components model to the agent-level outgoing $\text{Influence}(i)$, with intercept-only fixed effects and random intercepts for *agent* (nested within networks), *networks realization*, and *statement*. The model formula is

`influence ~ 1 + (1 | network-type) + (1 | statement) + (1 | agent/network-type)`

This allows us to quantify the variance in *outgoing influence* attributed to agent identity relative to graph realization and statement content.

For each random-effects component, we report the intraclass correlation coefficient (ICC) as the proportion of total variance attributable to that component. A higher agent-level ICC suggests the existence of persistent opinion leaders, whereas a near-zero value indicates that influence is largely exchangeable across agents (no stable opinion leaders). In Figure 4, we report 95% confidence intervals estimated via a parametric bootstrap (1,000 resamples).

To test whether the agent-level variance component is distinguishable from zero, we use an exact restricted likelihood ratio test (Scheipl et al., 2008), which accounts for the boundary problem in testing $\sigma_{\text{agent}}^2 = 0$. The null distribution is simulated with 10,000 Monte Carlo draws.

D.3 Contrast Analysis

We define four causal contrasts on the four scenarios (I, II, III and IV), each corresponding to a single-degree-of-freedom effect on the outcome. Table A6 specifies the contrast weights, and below we provide the contrast definitions:

Role Effect: Random Roles (II) vs. Baseline (I)

Model Effect: Random Specialists (III) vs. Random Roles (II)

Role-alignment: Matched Specialists (IV) vs. Random Specialists (III)

Heterogeneity: comparing the average of heterogeneous settings to the average of homogeneous settings.

Contrast	I. Baseline	II. Random Roles	III. Random Specialists	IV. Matched Specialists
Role effect	-1	+1	0	0
Model effect	0	-1	+1	0
Role-model alignment	0	0	-1	+1
Heterogeneity	-0.5	-0.5	+0.5	+0.5

Table A6: Contrast weights for the four causal contrasts on the scenario factors. Each row defines a linear combination of the four settings tested as a single-degree-of-freedom contrast. Resulting effect sizes are visualized in Figure A1.

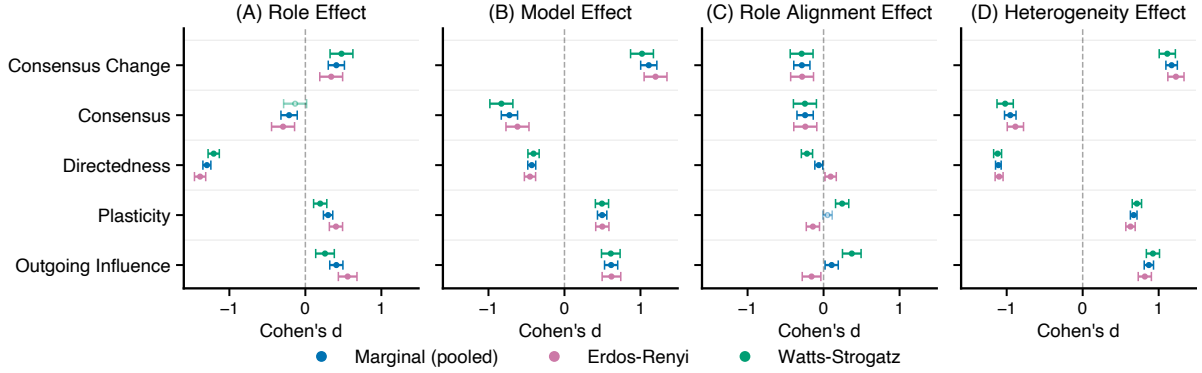


Figure A1: Effect sizes for the four scenario contrasts, shown as marginal estimates averaged over network types and conditional estimates within ER and WS networks. The results are provided in Table A9.

For each contrast, we compute both marginal and conditional within-networks estimates. The marginal contrasts average across network types, yielding overall *population-level* effects for each manipulation. The within-network (simple-effects) contrasts estimate the same linear combination separately for each network type, allowing us to test whether the effect of a given manipulation depends on the network realization. In Figure A1, we report the effect size (Cohen's d) associated with the manipulations.

E Dynamics Implementation

To complement the mixed-effects regression analysis, we fit a hierarchy of interpretable multinomial transition models to the empirical belief trajectories produced by the LLM-agent simulations. The goal of this analysis is descriptive rather than primarily predictive: we use the model hierarchy as an ablation framework to investigate which sources of information are necessary to reproduce the observed belief-update dynamics.

The hierarchy progressively incorporates information about: **(i)** individual persistence, **(ii)** global population composition (inspired by DeGroot (1974)), **(iii)** local neighborhood composition (inspired by Friedkin and Johnsen (1990)), and **(iv)** agent identity and assigned role information. Held-out predictive performance and rollout evaluation are used to assess whether these information sources capture stable structure in the observed dynamics.

Let $B_i^{(r,t)} \in \{-1, 0, 1\}$ denote the belief label of agent a_i in simulation run r at round t , where $-1, 0$, and 1 correspond to rejecting, abstaining/uncertain, and accepting the statement, respectively.

Each model estimates a conditional distribution over next-round beliefs,

$$\Pr(B_i^{(t+1)} = c \mid \phi_i^{(t)}), \quad c \in \{-1, 0, 1\},$$

where $\phi_i^{(t)}$ is a model-specific feature vector observed at round t .

Models are fit separately for each graph type, where $g \in \{\text{ER}, \text{WS}\}$, and experimental condition, where $e \in \{\text{I}, \text{II}, \text{III}, \text{IV}\}$. We pool over statements and random seeds within each $\langle g, e \rangle$ subset.

E.1 Model Hierarchy

Table A7 summarizes the hierarchy of multinomial transition models used in the analysis. The hierarchy functions as an ablation study over different sources of information available during belief updating.

M1: Persistence. M1 estimates empirical transition probabilities

$$\Pr(B_i^{(t+1)} = c \mid B_i^{(t)})$$

directly from observed transition frequencies. This model ignores the network, the global population state, and agent identity, and therefore serves as a baseline for individual-level persistence.

M2: Global drift + persistence. Let

$$\mathbf{g}^{(r,t)} = (f_{-1}^{(r,t)}, f_0^{(r,t)}, f_1^{(r,t)})$$

denote the fraction of agents in run r holding each belief state at round t . M2 then predicts next-round beliefs using this global belief distribution and an agent’s current belief:

$$\Pr(B_i^{(t+1)} = c \mid B_i^{(t)}, \mathbf{g}^{(t)}).$$

Comparisons between M1 and M2 therefore test whether global population composition contributes explanatory structure beyond individual persistence alone.

M3: Local social influence. M3 replaces global population features with local neighborhood composition features. Let

$$\mathbf{n}_i^{(r,t)} = (n_{i,-1}^{(r,t)}, n_{i,0}^{(r,t)}, n_{i,1}^{(r,t)})$$

denote the fraction of neighbors in $\mathcal{N}(i)$ holding each belief state at round t . Then, M3 estimates

$$\Pr(B_i^{(t+1)} = c \mid B_i^{(t)}, \mathbf{n}_i^{(t)}).$$

Improvements over M1 and M2 suggest that local neighborhood exposure contains explanatory information beyond persistence and global population state.

M4: Local social influence + identity. M4 extends M3 by adding categorical features for the agent’s underlying LLM and assigned role:

$$\Pr(B_i^{(t+1)} = c \mid B_i^{(t)}, \mathbf{n}_i^{(t)}, \mathcal{M}_i, \rho_i),$$

where $\mathcal{M}_i$ denotes the LLM assigned to agent a_i and ρ_i denotes its assigned role label. Improvements from M3 to M4 therefore indicate that LLM identity and assigned role explain additional heterogeneity in belief updating after accounting for persistence and local neighborhood exposure.

E.2 Fitting Procedure and Evaluation

For each model, graph type, and experimental condition, we construct transition-level examples from the empirical trajectories. Models are trained and evaluated using train, validation, and test splits defined at the run level, so that transitions from the same simulation run do not appear in multiple splits. This prevents leakage across temporally and structurally related observations within the same trajectory.

Model	Persistence	Global state	Neighborhood belief	Identity / role
M1: Persistence	✓	–	–	–
M2: Global drift + persistence	✓	✓	–	–
M3: Local social influence	✓	–	✓	–
M4: Local social influence + identity	✓	–	✓	✓

Table A7: **Hierarchy of multinomial transition models.** The hierarchy functions as a descriptive ablation study over information sources available during belief updating. M1 captures individual persistence, M2 combines persistence and global belief, M3 captures local neighborhood exposure, and M4 additionally incorporates LLM identity and assigned role information.

Held-out predictive evaluation. We first evaluate one-step predictive fit on held-out transitions. For each split, the fitted model predicts $\hat{B}_i^{(t+1)}$ from the relevant features at round t . Performance is computed by comparing predicted next-round beliefs to empirical next-round beliefs on the train, validation, and test splits using Matthews correlation coefficient (MCC).

Behavioral rollout evaluation. We additionally evaluate whether the fitted transition models reproduce collective behavioral dynamics when simulated forward. For each held-out run, we initialize the model from the empirical belief state at the starting round and iteratively simulate subsequent rounds using stochastic updates, where next-round beliefs are sampled from predicted transition probabilities.

We compare empirical and simulated trajectories using *consensus fidelity* by computing run-level absolute error between the consensus of the empirical and simulated trajectories.

These rollout comparisons evaluate whether the fitted transition rules reproduce not only one-step transitions, but also the collective behavioral patterns that emerge when the system is simulated forward.

E.3 Interpreting Model Comparisons

The primary object of interest in this analysis is not the individual coefficient values of the fitted multinomial models, but rather the comparative performance of the model hierarchy. Because the hierarchy progressively introduces additional sources of information, improvements between adjacent models can be interpreted as evidence that the newly introduced information source contributes explanatory structure to the observed dynamics.

For example, improvements from M1 to M2 indicate that global population composition contains information beyond individual persistence alone.

Improvements from M2 to M3 indicate that local neighborhood composition contains explanatory structure beyond global population state. Finally, improvements from M3 to M4 indicate that LLM identity and assigned role are associated with systematic heterogeneity in belief updating even after accounting for persistence and local exposure.

Accordingly, the hierarchy is interpreted primarily as a descriptive ablation framework over information sources rather than as a search for a single best predictive model.

F Tables

In this section, we report detailed tables of the contrast analysis: marginal effect size in Table A8 and conditional effect on the network structure in Table A9.

Outcome	Contrast	Cohen's d	CI _{.025}	CI _{.975}	SE	t -stat	p -val	df
Plasticity	Role Effect	0.04	0.03	0.04	0.00	9.55	<0.001	1253.00
	Model Effect	0.06	0.05	0.07	0.00	15.83	<0.001	1253.00
	Role Alignment	0.01	-0.00	0.01	0.00	1.66	0.098	1253.00
	Heterogeneity	0.08	0.07	0.08	0.00	30.30	<0.001	1253.00
Directedness	Role Effect	-0.49	-0.51	-0.47	0.01	-48.59	<0.001	1253.00
	Model Effect	-0.16	-0.18	-0.14	0.01	-16.14	<0.001	1253.00
	Role Alignment	-0.02	-0.04	-0.00	0.01	-2.35	0.019	1253.00
	Heterogeneity	-0.42	-0.43	-0.41	0.01	-58.85	<0.001	1253.00
Outgoing Influence	Role Effect	0.04	0.03	0.04	0.00	9.23	<0.001	1253.00
	Model Effect	0.06	0.05	0.06	0.00	13.91	<0.001	1253.00
	Role Alignment	0.01	0.00	0.02	0.00	2.41	0.016	1253.00
	Heterogeneity	0.08	0.07	0.09	0.00	27.90	<0.001	1253.00
Consensus Change	Role Effect	0.05	0.04	0.07	0.01	7.52	<0.001	1253.00
	Model Effect	0.14	0.13	0.15	0.01	20.46	<0.001	1253.00
	Role Alignment	-0.04	-0.05	-0.02	0.01	-5.27	<0.001	1253.00
	Heterogeneity	0.15	0.14	0.16	0.01	30.53	<0.001	1253.00
Consensus	Role Effect	-0.02	-0.04	-0.01	0.01	-3.92	<0.001	1238.97
	Model Effect	-0.08	-0.09	-0.07	0.01	-13.27	<0.001	1238.97
	Role Alignment	-0.03	-0.04	-0.01	0.01	-4.45	<0.001	1238.97
	Heterogeneity	-0.10	-0.11	-0.10	0.00	-24.69	<0.001	1238.97

Table A8: Marginal contrasts on key outcomes (Appendix D). Cohen's d specifies the effect size (with 95% Confidence Intervals). The contrasts include *role effect* (contrast between scenarios I and II), *model effect* (contrast between scenarios III and II), *role alignment* (contrast between scenarios IV and III), and *heterogeneity* (contrast between III+IV and I+II). A high effect size indicates that outcomes differ between scenarios. The sign of d shows the direction of the effect. SE is the standard error for the d , t -stat is a t -statistic, p -value shows the significance of the d , df stands for degrees of freedom.

Outcome	Network	Contrast	Cohen's d	CI _{.025}	CI _{.975}	SE	t -stat	p -val	df
Plasticity	ER	Role Effect	0.05	0.04	0.06	0.01	9.09	<0.001	1253.00
		Model Effect	0.06	0.05	0.07	0.01	11.24	<0.001	1253.00
		Role Alignment	-0.02	-0.03	-0.01	0.01	-3.17	0.002	1253.00
		Heterogeneity	0.07	0.07	0.08	0.00	20.09	<0.001	1253.00
	WS	Role Effect	0.02	0.01	0.03	0.01	4.41	<0.001	1253.00
		Model Effect	0.06	0.05	0.07	0.01	11.14	<0.001	1253.00
		Role Alignment	0.03	0.02	0.04	0.01	5.52	<0.001	1253.00
		Heterogeneity	0.08	0.08	0.09	0.00	22.77	<0.001	1253.00
Directedness	ER	Role Effect	-0.53	-0.55	-0.50	0.01	-36.73	<0.001	1253.00
		Model Effect	-0.17	-0.20	-0.14	0.01	-12.03	<0.001	1253.00
		Role Alignment	0.04	0.01	0.06	0.01	2.47	0.014	1253.00
		Heterogeneity	-0.42	-0.44	-0.40	0.01	-41.24	<0.001	1253.00
	WS	Role Effect	-0.46	-0.49	-0.43	0.01	-31.99	<0.001	1253.00
		Model Effect	-0.15	-0.18	-0.13	0.01	-10.79	<0.001	1253.00
		Role Alignment	-0.08	-0.11	-0.06	0.01	-5.80	<0.001	1253.00
		Heterogeneity	-0.42	-0.45	-0.41	0.01	-41.98	<0.001	1253.00
Outgoing Influence	ER	Role Effect	0.05	0.04	0.06	0.01	8.92	<0.001	1253.00
		Model Effect	0.06	0.04	0.07	0.01	9.90	<0.001	1253.00
		Role Alignment	-0.01	-0.03	-0.00	0.01	-2.53	0.011	1253.00
		Heterogeneity	0.07	0.07	0.08	0.00	18.52	<0.001	1253.00
	WS	Role Effect	0.02	0.01	0.04	0.01	4.14	<0.001	1253.00
		Model Effect	0.06	0.04	0.07	0.01	9.77	<0.001	1253.00
		Role Alignment	0.03	0.02	0.04	0.01	5.94	<0.001	1253.00
		Heterogeneity	0.09	0.08	0.09	0.00	20.94	<0.001	1253.00
Consensus Change	ER	Role Effect	0.04	0.02	0.06	0.01	4.44	<0.001	1253.00
		Model Effect	0.15	0.13	0.17	0.01	15.63	<0.001	1253.00
		Role Alignment	-0.04	-0.06	-0.02	0.01	-3.69	<0.001	1253.00
		Heterogeneity	0.16	0.14	0.17	0.01	22.64	<0.001	1253.00
	WS	Role Effect	0.06	0.04	0.08	0.01	6.20	<0.001	1253.00
		Model Effect	0.13	0.11	0.15	0.01	13.31	<0.001	1253.00
		Role Alignment	-0.04	-0.06	-0.02	0.01	-3.77	<0.001	1253.00
		Heterogeneity	0.14	0.13	0.15	0.01	20.54	<0.001	1253.00
Consensus	ER	Role Effect	-0.03	-0.05	-0.01	0.01	-3.80	<0.001	1238.97
		Model Effect	-0.07	-0.08	-0.05	0.01	-8.01	<0.001	1238.97
		Role Alignment	-0.03	-0.04	-0.01	0.01	-3.12	0.002	1238.97
		Heterogeneity	-0.10	-0.11	-0.08	0.01	-16.22	<0.001	1238.97
	WS	Role Effect	-0.01	-0.03	0.00	0.01	-1.75	0.081	1238.97
		Model Effect	-0.09	-0.11	-0.07	0.01	-10.76	<0.001	1238.97
		Role Alignment	-0.03	-0.04	-0.01	0.01	-3.18	0.002	1238.97
		Heterogeneity	-0.11	-0.12	-0.10	0.01	-18.69	<0.001	1238.97

Table A9: Conditional contrasts on key outcomes (Appendix D). Cohen's d specifies the effect size (95% Confidence Intervals). Here we stratify the contrast based on the network structure: ER stands for Erdős–Rényi, and WS stands for Watts-Strogatz. The contrasts include *role effect* (contrast between scenarios I and II), *model effect* (contrast between scenarios III and II), *role alignment* (contrast between scenarios IV and III), and *heterogeneity* (contrast between III+IV and I+II). A high effect size indicates that outcomes differ between scenarios. The sign of d shows the direction of the effect. SE is the standard error for the d , t -stat is a t -statistic, p -value shows the significance of the d , df stands for degrees of freedom.